

Markscheme

Mathematics: analysis and approaches
Practice question bank

100. (a)

The power rule $\frac{d}{dx}(x^n) = nx^{n-1}$ can be proven by induction for every positive integer n , using the product rule and the base case $n = 1$.

Using this result, find the value of $\frac{d}{dx}(x^7)$ at $x = 2$.

$$\frac{d}{dx}(x^7) = 7x^6 \quad (M1)$$

$$\text{substitute } x = 2: 7(2^6) \quad (M1)$$

$$= 7(64) \quad \mathbf{A1}$$

$$= 448 \quad \mathbf{A1}$$

[6 marks]